Power Series and Taylor Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Taylor and Maclaurin series

Find the first three nonzero terms of the Maclaurin series for $f(x) = e^x$, using $f^{(n)}(0) = 1$ for all n.

$e^x = 1 + x + \frac{x^2}{2!} + \cdots = 1 + x + \frac{x^2}{2} + \cdots$.

2 [Foundational] • Target: Taylor and Maclaurin series

Find the third-degree Taylor polynomial for $f(x) = \cos x$ centered at $x = 0$.

$f(0) = 1$, $f'(0) = 0$, $f''(0) = -1$, $f'''(0) = 0$: $T_3(x) = 1 - \frac{x^2}{2}$.

3 [Foundational] • Target: Taylor and Maclaurin series

Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots$ to approximate $\sin(0.2)$ using the first two nonzero terms, to four decimal places.

$0.2 - \frac{(0.2)^3}{6} = 0.2 - 0.001333\ldots \approx 0.1987$.

4 [Foundational] • Target: Interval of convergence (including endpoints)

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$, checking both endpoints.

Ratio test gives radius $R = 1$, centered at $x = 2$: $1 < x < 3$ tentatively. At $x = 3$: $\sum \frac{1}{n}$ diverges (harmonic). At $x = 1$: $\sum \frac{(-1)^n}{n}$ converges (alternating). Interval of convergence: $[1, 3)$.

5 [Foundational] • Target: Using known series

Using the known Maclaurin series $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, write the Maclaurin series for e^{-x^2} (substitute $-x^2$ for x), giving the first three terms.

$e^{-x^2} = 1 + (-x^2) + \frac{(-x^2)^2}{2!} + \cdots = 1 - x^2 + \frac{x^4}{2} - \cdots$.

6 [Foundational] • Target: Using known series

Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, write the power series for $\frac{1}{1+x^2}$ (substitute $-x^2$ for x).

$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n} = 1 - x^2 + x^4 - x^6 + \cdots$.

7 [Foundational] • Target: Taylor series centered at a nonzero point

Find the first three terms of the Taylor series for $f(x) = \ln x$ centered at $x = 1$, using $f(1) = 0$, $f'(1) = 1$, $f''(1) = -1$.

$T(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \cdots = (x-1) - \frac{(x-1)^2}{2} + \cdots$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Taylor series centered at a nonzero point**

Find the second-degree Taylor polynomial for $f(x) = \sqrt{x}$ centered at $x = 4$, given $f(4) = 2$, $f'(4) = \frac{1}{4}$, $f''(4) = -\frac{1}{32}$.

$T_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$.

9 **[Intermediate]** • **Target: Error estimation with Taylor polynomials**

Using the second-degree Maclaurin polynomial for $\cos x$, $T_2(x) = 1 - \frac{x^2}{2}$, estimate $\cos(0.3)$ to four decimal places, and compare to the true value $\cos(0.3) \approx 0.9553$.

$T_2(0.3) = 1 - \frac{0.09}{2} = 1 - 0.045 = 0.9550$, very close to the true value 0.9553 (error ≈ 0.0003).

10 **[Intermediate]** • **Target: Visualizing a Taylor approximation**

The graph compares $f(x) = \cos x$ to its second-degree Maclaurin polynomial $T_2(x) = 1 - \frac{x^2}{2}$. Use the graph to describe where the approximation is best and where it breaks down.

The two curves are nearly indistinguishable near $x = 0$, where the approximation is best (the region of the expansion point). As $|x|$ increases toward $\pm\pi$, $T_2(x)$ (a downward parabola) diverges sharply from the bounded, oscillating $\cos x$, showing the approximation breaks down away from the center.

11 **[Intermediate]** • **Target: Differentiating and integrating power series**

Given $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $|x| < 1$, differentiate both sides term-by-term to find a power series for $\frac{1}{(1-x)^2}$.

$\frac{d}{dx}\left[\frac{1}{1-x}\right] = \frac{1}{(1-x)^2}$. Differentiating the series: $\sum_{n=1}^{\infty} n x^{n-1}$. So $\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1} = 1 + 2x + 3x^2 + \cdots$.

12 **[Intermediate]** • **Target: Differentiating and integrating power series**

Using the Maclaurin series $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$, integrate term-by-term to find the first three terms of the power series for $\int \sin x\, dx = -\cos x + C$, and confirm they match the known Maclaurin series for $-\cos x$.

$\int \left(x - \frac{x^3}{6} + \cdots\right) dx = \frac{x^2}{2} - \frac{x^4}{24} + \cdots + C$. Comparing to $-\cos x = -1 + \frac{x^2}{2} - \frac{x^4}{24} + \cdots$: matching term-by-term (aside from the constant -1, absorbed into C) confirms consistency.

13 **[Intermediate]** • **Target: Using Taylor series to evaluate a limit**

Use the Maclaurin series $\sin x = x - \frac{x^3}{6} + \cdots$ to evaluate $\lim_{x \to 0} \frac{\sin x - x}{x^3}$.

$\sin x - x = -\frac{x^3}{6} + \frac{x^5}{120} - \cdots$, so $\frac{\sin x - x}{x^3} = -\frac{1}{6} + \frac{x^2}{120} - \cdots \to -\frac{1}{6}$ as $x \to 0$.

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14 [Intermediate] • Target: Using Taylor series to evaluate a limit

Find the fourth-degree Taylor polynomial for $f(x) = e^x$ centered at $x = 0$, and use it to estimate $e^{0.5}$, comparing to the true value $e^{0.5} \approx 1.6487$.

$T_4(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}$. $T_4(0.5) = 1 + 0.5 + 0.125 + 0.02083 + 0.0026 \approx 1.6484$, very close to the true value 1.6487.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Using Taylor series to evaluate a limit • Cross-Lesson: 7.02 Integration by Parts

An engineer needs a quick approximation for $\ln(1.1)$ without a calculator. Using the Maclaurin series $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ for $|x| < 1$, find the approximation using the first three terms with $x = 0.1$, and compare to the true value $\ln(1.1) \approx 0.09531$.

$\ln(1.1) \approx 0.1 - \frac{(0.1)^2}{2} + \frac{(0.1)^3}{3} = 0.1 - 0.005 + 0.000333 \approx 0.095333$, extremely close to the true value 0.09531 (error ≈ 0.00002).

16 [Challenging] • Target: Visualizing a Taylor approximation • Cross-Lesson: 8.01 Area Between Curves

Level 3 · Advanced Find the full interval of convergence (including endpoint behavior) for the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n^2 \cdot 4^n}$.

Apply the Ratio Test: $L = \lim_{n\to\infty} \left| \frac{(2x-1)^{n+1}}{(n+1)^2 4^{n+1}} \cdot \frac{n^2 4^n}{(2x-1)^n} \right| = \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^2 \cdot \frac{|2x-1|}{4} = \frac{|2x-1|}{4}$ Require x: $2x - 1 = 4$, giving $\sum_{n=1}^{\infty} \frac{4^n}{n^2 4^n} = \sum_{n=1}^{\infty} \frac{1}{n^2}$, which converges by the p-series test ($p = 2 > 1$). Endpoint $x = -3/2$: $2x - 1 = -4$, giving.

17 [Challenging] • Target: Differentiating and integrating power series • Cross-Lesson: 8.02 Volumes of Revolution

[Convergence Tests & Power Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

18 [Challenging] • Target: Using Taylor series to evaluate a limit • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Using the binomial series for $(1 - x^2)^{-1/2}$, derive the Maclaurin series expansion for $f(x) = \arcsin x$ up to and including the x^5 term. State its radius of convergence.

1. Recall the general binomial expansion $(1+u)^p = 1 + pu + \frac{p(p-1)}{2!}u^2 + \ldots$. Here $u = -x^2$ and $p = -1/2$: $(1-x^2)^{-1/2} = 1 + (-\frac{1}{2})(-x^2) + \frac{(-1/2)(-3/2)}{2}(-x^2)^2 + O(x^6) = 1 + \frac{1}{2}x^2 + \frac{3}{8}x^4 + O(x^6)$. 2. Since $\arcsin x = \int_0^x \frac{1}{\sqrt{1-t^2}}\,dt$, integrate term-by-term: $\arcsin x = \int_0^x (1 + \frac{1}{2}t^2 + \frac{3}{8}t^4 + \ldots)\,dt = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \ldots$ 3. The binomial series converge.

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19 [Challenging] • Target: Radius and interval of convergence • Cross-Lesson: 9.01 Separable Differential Equations

Level 1 · FoundationsFind the radius of convergence R for the power series $\sum_{n=0}^{\infty} n!\,(2x-1)^n$.

Apply the Ratio Test: $L = \lim_{n\to\infty}\left|\frac{(n+1)!\,(2x-1)^{n+1}}{n!\,(2x-1)^n}\right| = |2x-1|\lim_{n\to\infty}(n+1)$. For any $x \neq \frac{1}{2}$, this limit is ∞. The series converges only at its center $x=\frac{1}{2}$. Thus, the radius of convergence is **R = 0**.

20 [Challenging] • Target: Taylor and Maclaurin series • Cross-Lesson: 1.01 Precalculus Review for Calculus

Level 1 · FoundationsExpress the infinite summation limit as a definite integral on the interval $[0,1]$ and evaluate it: $\lim_{n\to\infty}\sum_{i=1}^{n}\frac{i^3}{n^4}$.

Factor $\frac{1}{n}=\Delta x$: $\sum_{i=1}^{n}\frac{i^3}{n^4} = \sum_{}^{n}\left(\frac{i}{n}\right)^3\cdot\frac{1}{n}$. Here $x_i=\frac{i}{n}=0+i\Delta x$ with $\Delta x=\frac{1}{n}$, matching the right Riemann sum of $f(x)=x^3$ on $[0,1]$: $\lim_{n\to\infty}\sum_{i=1}^{}\left(\frac{i}{n}\right)^3\frac{1}{n}=\int_0^1 x^3\,dx=\left[\frac{x^4}{4}\right]_0^1=\frac{1^4}{4}-0=\mathbf{\frac{1}{4}}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Radius and interval of convergence

Interval of convergence of $\sum\frac{(-3)^n(x-2)^n}{\sqrt{n}}$: centre, radius, and endpoints:

a. $R=1/3$**, interval** $(5/3,\,7/3]$ ✓

b. $R=1/3$, interval $[5/3,\,7/3)$

c. $R=1/3$, interval $[5/3,\,7/3]$

d. $R=3$, interval $(-1,\,5]$

$|-3(x-2)|<1\Rightarrow|x-2|<1/3$. At $x=7/3$: $\sum(-1)^n/\sqrt{n}$ converges (Leibniz). At $x=5/3$: $\sum 1/\sqrt{n}$ diverges.

22 [Foundational] • Target: Taylor and Maclaurin series

The interval of convergence always contains at least the point:

a. $x=R$

b. $x=1$

c. $x=-1$

d. $x=0$ ✓

Series converges at its centre; if centred at 0, converges at $x=0$.

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23 [Foundational] • Target: Interval of convergence (including endpoints)

The error in using $T_n(x)$ decreases as n increases (assuming convergence). This is because the remainder $R_n \to$

a. 1

b. $f(x)$

c. $n!$

d. 0 ✓

Remainder $R_n \to 0$ for convergent Taylor series.

24 [Foundational] • Target: Using known series

Maclaurin series for $\tan^{-1} x$ at $x = 1$ gives $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \cdots$. This converges very slowly. The 10th partial sum approximates $\pi/4$ with error less than:

a. 1/11

b. 1/21 ✓

c. 1/19

d. 0.01

Alternating series: error $\leq a_{11} = 1/21$.

25 [Foundational] • Target: Taylor series centered at a nonzero point

If $|f^{(4+1)}(c)| \leq 1$ on the relevant interval, bound the Lagrange error $|R_4(x)|$ for approximating f at $x = 0.1$ using a Taylor polynomial centred at $a = 1$.

a. 0.0098415

b. **0.00492075** ✓

c. 0.00246038

d. 0.0273375

Lagrange error bound: $|R_4(x)| \leq \frac{M|x-a|^{4+1}}{(4+1)!} = \frac{1 \cdot 0.9^5}{5!} \approx 0.00492075$.

26 [Foundational] • Target: Error estimation with Taylor polynomials

Composite series: $e^{\sin x}$ up to x^3. Using $\sin x = x - x^3/6 + \cdots$ inside $e^u = 1 + u + u^2/2 + \cdots$:

a. $1 + x + \frac{x^2}{2}$ ✓

b. e

c. $1 + x + \frac{x^2}{2} + \frac{x^3}{6}$

d. $1 + x + \frac{x^2}{2} - \frac{x^3}{3}$

Substituting $u = x - \frac{x^3}{6} + \ldots$ into $e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + \ldots$ gives $1 + \left(x - \frac{x^3}{6}\right) + \frac{x^2}{2} + \frac{x^3}{6} + O(x^4) = 1 + x + \frac{x^2}{2}$.

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27 [Foundational] • Target: Visualizing a Taylor approximation

Taylor series of $1/(1-x)$ at $x=0$:

a. $\sum x^n/n!$

b. $\sum(-1)^n x^n$

c. $\sum x^{n+1}$

d. $\sum x^n$ **for** $|x|<1$ ✓

Applying Convergence Tests and Power Series principles, Geometric series. Therefore, the verified correct option is $\sum x^n$ for $|x|<1$.

28 [Foundational] • Target: Differentiating and integrating power series

Use alternating series estimation: $|\cos 1 - 1 + 1/2| \le |-1/4!\,| =$

a. $1/12$

b. $1/120$

c. $1/6$

d. $1/24$ ✓

Applying Convergence Tests and Power Series principles, Error $\le a_4 = 1/4! = 1/24$. Therefore, the verified correct option is $1/24$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Using Taylor series to evaluate a limit

Error bound: $|\cos x - 1 + x^2/2| \le x^4/24$ (from T_4 bound). This means the error at $x=0.1$ is at most:

a. $0.0001/24$

b. $0.1^4/24$ ✓

c. $0.1^2/2$

d. $1/24$

Applying Convergence Tests and Power Series principles, $|x^4/24| = 0.0001/24$. Therefore, the verified correct option is $0.1^4/24$.

30 [Intermediate] • Target: Radius and interval of convergence

Radius of convergence of $\sum x^n/n$ is $R=$

a. 0

b. e

c. ∞

d. 1 ✓

Applying Convergence Tests and Power Series principles, $R = \lim n/(n+1) = 1$. Therefore, the verified correct option is 1.

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31 [Intermediate] • Target: Taylor and Maclaurin series

A power series $\sum c_n x^n$ with radius of convergence R converges absolutely for:

a. no x

b. all x

c. **|x|<R ✓**

d. $|x| > R$

By definition of the radius of convergence, the series converges absolutely inside $(-R, R)$ and diverges outside $[-R, R]$.

32 [Intermediate] • Target: Interval of convergence (including endpoints)

The Maclaurin series for $\sqrt{1+x}$ is a special case of the binomial series with $\alpha =$

a. 2

b. 1

c. $-1/2$

d. $1/2$ ✓

Applying Convergence Tests and Power Series principles, $(1+x)^{1/2}$. Therefore, the verified correct option is $1/2$.

33 [Intermediate] • Target: Using known series

Term-by-term differentiation is valid inside the radius of convergence. The derivative of $\sum_{n=0}^{\infty} x^n/(n+1)$ is:

a. $\sum x^n/n$

b. $\sum n x^{n-1}/(n+1)$

c. $1/(1-x)$ ✓

d. $\ln(1-x)$

Recognise $\int \frac{1}{1-x} dx = -\ln(1-x) = \sum x^n/(n+1)$; differentiate $\to 1/(1-x)$.

34 [Intermediate] • Target: Taylor series centered at a nonzero point

Approximate $\int_0^1 \frac{\sin x}{x} dx$ using power series (first three terms):

a. $1 - 1/3 + 1/10$

b. $1 - 1/18 + 1/600$ ✓

c. $1 - 1/6 + 1/20$

d. $1 - 1/18$

$\sin x/x = \sum (-1)^n x^{2n}/(2n+1)!$; $\int_0^1 = \sum (-1)^n/(2n+1)(2n+1)! = 1 - 1/18 + 1/600 - \cdots$.

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35 [Intermediate] • Target: Error estimation with Taylor polynomials

Find radius of convergence: $\sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2} x^n$:

a. $1/4$ ✓

b. $1/2$

c. ∞

d. 4

Central binomial: $\binom{2n}{n} x^n$; $|c_n|^{1/n} \to 4 \Rightarrow R = 1/4$.

36 [Intermediate] • Target: Visualizing a Taylor approximation

If the radius of convergence of $\sum a_n x^n$ is R, then the radius of $\sum a_n x^{2n}$:

a. R

b. R^2

c. $\sqrt{R}$ ✓

d. $2R$

Substitute $y = x^2$: $|y| < R \Rightarrow |x|^2 < R \Rightarrow |x| < \sqrt{R}$.

37 [Intermediate] • Target: Differentiating and integrating power series

The Maclaurin series for e^{-x^2} (substitute $-x^2$ for x in e^x series):

a. $\sum \frac{(-1)^n x^{2n}}{n!}$ ✓

b. $\sum x^{2n}/n!$

c. $\sum (-x)^n/n!$

d. $\sum x^n/n!$

$e^{-x^2} = \sum \frac{(-x^2)^n}{n!} = \sum \frac{(-1)^n x^{2n}}{n!}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Using Taylor series to evaluate a limit • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Sum $\sum_{n=1}^{\infty} n(n-1)x^n/(n!)$ simplifies using e^x series to:

a. $e^x(x^2 - x)$

b. xe^x

c. $x^2 e^x$ ✓

d. $x^2 e^x/2$

$n(n-1)/n! = 1/(n-2)! \Rightarrow x^2 \sum x^{n-2}/(n-2)! = x^2 e^x$.

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39 [Challenging] • Target: Radius and interval of convergence • Cross-Lesson: 9.01 Separable Differential Equations

The power series $\sum a_n x^n$ represents a function that is C^∞ (infinitely differentiable) inside its radius of convergence. This means all derivatives:

a. are polynomials

b. are zero

c. **exist** ✓

d. diverge

Power series define smooth (C^∞) functions inside R.

40 [Challenging] • Target: Taylor and Maclaurin series • Cross-Lesson: 1.01 Precalculus Review for Calculus

Power series $\sum \frac{(x-2)^n}{3^n}$: centre and radius:

a. centre 0, $R = 3$

b. centre 3, $R = 2$

c. centre 2, $R = 1/3$

d. **centre 2,** $R = 3$ ✓

Centre $x = 2$, ratio $r = |(x-2)/3| < 1 \Rightarrow R = 3$.

41 [Challenging] • Target: Interval of convergence (including endpoints) • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Using power series, compute $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$:

a. $1/2$ ✓

b. 1

c. 0

d. 2

$1 - \cos x \approx x^2/2 \Rightarrow (x^2/2)/x^2 = 1/2$.

42 [Challenging] • Target: Using known series • Cross-Lesson: 11.01 Sequences and Series

Power series $\sum_{n=0}^{\infty} n(x-1)^n$. Radius of convergence:

a. 0

b. 1 ✓

c. 2

d. ∞

$L = \lim n^{1/n} \cdot |x-1| \to |x-1|; R = 1$.

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43 [Challenging] • Target: Taylor series centered at a nonzero point • Cross-Lesson: 12.01 Convergence Tests

Power series centred at a: the interval of convergence always includes:

a. $(-\infty, \infty)$

b. $x = a$ ✓

c. the endpoint a

d. $x = 0$

Power series always converges at its centre $x = a$.

44 [Challenging] • Target: Error estimation with Taylor polynomials • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Hypergeometric series ${}_2F_1(a, b; c; x) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n n!} x^n$ (Pochhammer symbol). Radius of convergence:

a. depends on a, b, c

b. always ∞

c. **always 1** ✓

d. always 0

Gauss hypergeometric: radius of convergence 1.

45 [Challenging] • Target: Visualizing a Taylor approximation • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

If a function equals its Taylor series everywhere, it is called:

a. **analytic** ✓

b. polynomial

c. differentiable

d. continuous

Analytic means equal to its power series.

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